Hyperspectral Remote Sensing

IRS Chapter 14

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CSE620B

Quiz references: **Hyperspectral Remote Sensing Fundamentals** (Week 10).

Introduction to Hyperspectral Remote Sensing

Preview: we next connect narrow bands to spectroscopy, then examine calibration and AVIRIS. Image cubes, libraries, and unmixing follow in the next deck.

- **Hyperspectral** vs. conventional remote sensing
- Uses **many narrow spectral channels** (200+)
- Channels as narrow as **10 nm**
- Enables **fine spectral resolution**

Spectroscopy and Hyperspectral Data

- **Spectroscopy** enhances material identification
- Uses **detailed spectral data** for analysis
- Enables **matching spectra to libraries**
- Corrects atmospheric effects using internal data

Origins of Spectroscopy

- **Isaac Newton**: Separated light with prisms
- **William Wollaston**: Observed dark lines in spectra
- **Joseph Fraunhofer**: Identified solar absorption lines
- **Niels Bohr**: Linked spectral lines to atomic structure

Instruments in Spectroscopy

- **Spectroscope**: Basic instrument for viewing spectra
- **Spectrometer**: Measures light intensity at each wavelength
- **Spectrograph**: Records spectra over a range of wavelengths

Hyperspectral Imaging Applications

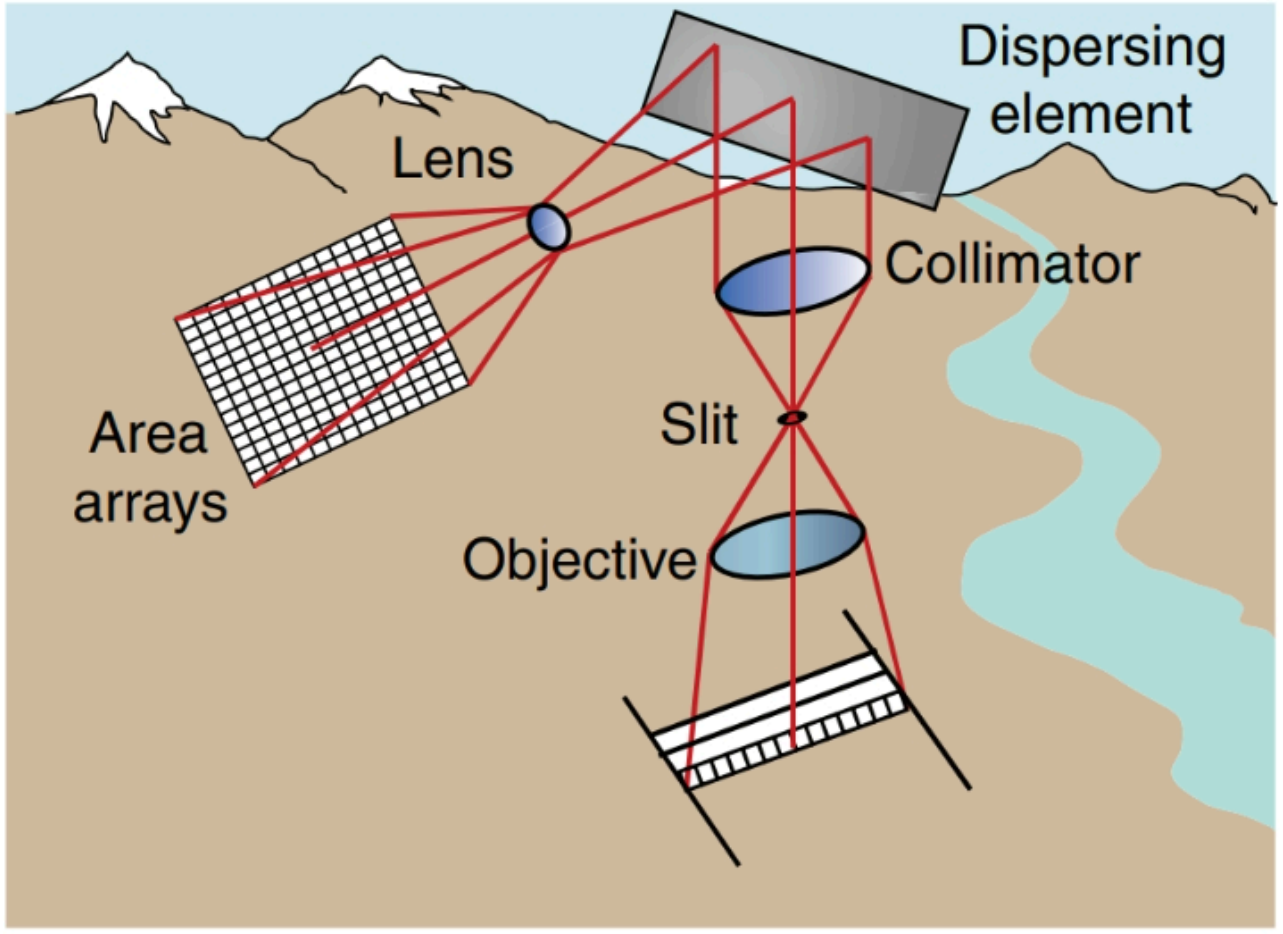
- Focuses on **identifying materials** through spectral signatures
- Extends beyond visible wavelengths to **NIR and SWIR**
- Used for **earth surface analysis** and atmospheric correction

Importance of Calibration

- Calibration **ensures data accuracy** across channels
- Essential for **consistent spectral measurements**
- Addresses **instrument and environmental variability**

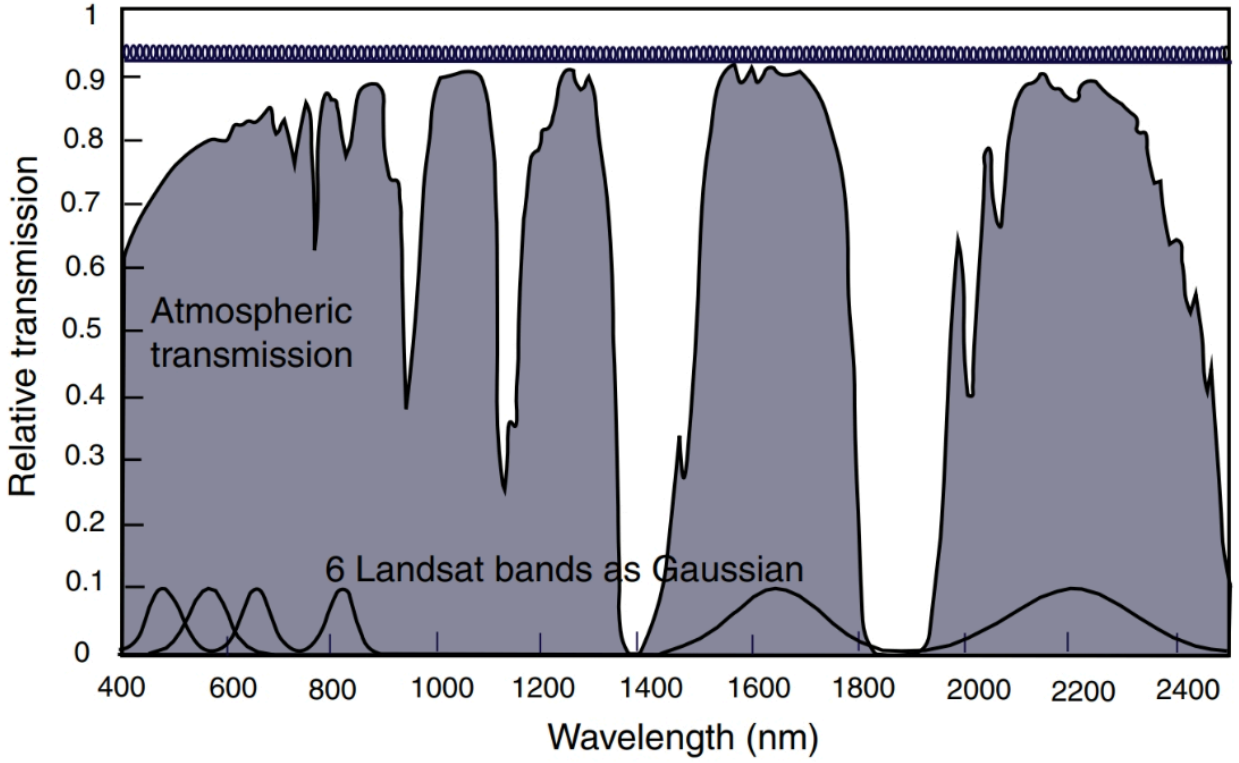
AVIRIS: A Key Hyperspectral Instrument

- Developed by **NASA/JPL** in the 1980s
- Collects **224 channels** across 400–2500 nm
- Enables **high-resolution hyperspectral imaging**



AVIRIS Spectral and Spatial Range

- Spectral range: **400–2500 nm**
- Channel width: **10 nm**
- Spatial resolution varies by altitude:
  - High altitude: ~**20 m** per pixel



Hyperspectral Data and Material Analysis

- Enables **precise material matching** to spectral libraries
- Common in **mineralogy, agriculture, and environmental studies**
- Can correct for **atmospheric effects**

Quiz Questions 13, 15

Classical Spectroscopy vs. Hyperspectral Imaging

- **Classical spectroscopy**: Analyzes single-point targets
- **Hyperspectral imaging**: Captures extensive spatial areas
- Applied to **large surface areas** rather than isolated points

Components of Hyperspectral Sensors

- **Diffraction grating**: Separates light into spectral bands
- **Detector arrays**: Capture specific spectral ranges (e.g., visible, NIR, SWIR)
- Fine **spectral, spatial, and radiometric resolution**

